



# Python for Fixed Income

COURSE 6 · NEURAL NETWORKS

## One-Dimensional Convolutions and Many Tenors

WEEK 3 · DAY 3

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# Three ways to read a window

## Dense, day 11

Every day its own weight: the window laid flat.

## Recurrent, day 12

Day by day, a state carried forward.

## Convolution, today

**Five weights slid along the window.**

Past tense: the 10 Yr change in bp on a date, from the 20 trading days before it.

## Allowed names

fitted change in bp

walk-forward error

the last value carried forward

## Never called

a forecast or a prediction

tomorrow's move or a call

an outlook or a view on rates

# A filter sliding over time



20 daily changes of the 10 Yr before 2015-02-03; a filter of width 5 covers days 1 to 5, then 2 to 6, ...

```
output[t] = sum of window[t + k] * weight[k] for k = 0 to 4, plus the bias  
outputs: 20 - 5 + 1 = 16          kernel shape (width, channels, filters): (5, 1, 4)
```

**Five weights of 0.2 are a 5-day moving average**

First output -2.8 bp; equal to pandas' `rolling(5).mean()` within 1e-12.

**A network fits its own weights: each filter is a moving sum with weights of its choosing.**

# A filter is a weighted moving sum

|            | Day 1 | Day 2   | Day 3  | Day 4  | Day 5   |
|------------|-------|---------|--------|--------|---------|
| Change, bp | -8.0  | -7.0    | -1.0   | 7.0    | -5.0    |
| Weight     | 0.351 | -0.4134 | 0.0233 | 0.4566 | -0.4714 |

## Same five weights at every step

Day 11's dense network gave each day its own weight; a filter has 5 however long the window.

Products added, plus a bias of 0: 5.6165

## 11 tenors as channels: kernel (5, 11, 4)

A filter is a block of 5 days by 11 tenors, 55 weights. Each output is one SUMPRODUCT over a rectangle; the output is still 16 steps per filter.

Keras computes a cross-correlation: the kernel is not flipped.

# Pooling: one number per filter

**7 of 16**

filter 1's outputs set to 0 by ReLU

**2.8086**

average pooling, filter 1

**9.5347**

max pooling, filter 1

```
keras.layers.Conv1D(16, 5, activation="relu") # (20, 1) -> (16, 16)
keras.layers.GlobalAveragePooling1D()        # (16, 16) -> (16,)
keras.layers.Dense(1)                        # (16,) -> 1, a change in either sign
```

**113**

weights and biases, one tenor

**913**

weights and biases, 11 tenors

# Walk-forward: the Conv1D on the 10 Yr

| Model, 10 Yr alone | MAE, bp | Folds below zero change |
|--------------------|---------|-------------------------|
| zero change        | 3.21    | -                       |
| training mean      | 3.22    | -                       |
| dense, day 11      | 3.35    | 0 of 5                  |
| SimpleRNN, day 12  | 3.31    | 0 of 5                  |
| LSTM, day 12       | 3.23    | 0 of 5                  |
| GRU, day 12        | 3.22    | 1 of 5                  |
| Conv1D, today      | 3.24    | 0 of 5                  |

## Best epoch by fold, Conv1D

1, 50, 1, 1, 50

Fitted changes spread 0.25 bp against the actual 4.33 bp: little to fit, so it stays near the training mean.

Same five folds, same settings, reseeded every fold: days 11 and 12 rebuilt here.

# Many tenors as channels

| Network | 10 Yr alone, bp | 11 tenors, bp | Folds below zero change |
|---------|-----------------|---------------|-------------------------|
| dense   | 3.35            | 3.87          | 0 of 5                  |
| LSTM    | 3.23            | 3.23          | 1 of 5                  |
| Conv1D  | 3.24            | 3.26          | 0 of 5                  |

Zero change on the same folds: 3.21 bp.

The 11-tenor Conv1D is not bit-identical run to run in its last digits (its MAE agreed to 6 decimals): printed as an MAE to 2 decimals and fold counts only.

**The flattened dense network got worse; the shared-weight networks barely moved.**

# Same-day changes stay out

| 10 Yr change against | Same date | One date earlier |
|----------------------|-----------|------------------|
| 2 Yr change          | 0.75      | 0.02             |
| 30 Yr change         | 0.93      | -0.03            |

## Known only after the day

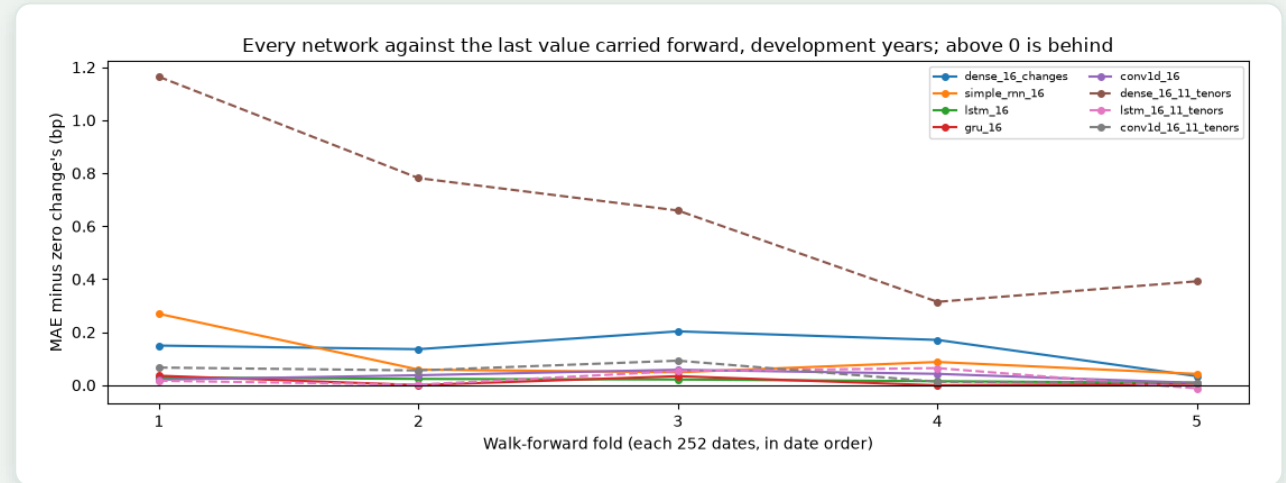
A window that ends on its target date holds other tenors' same-day changes.

```
netkit.assert_windows_past_only(changes.index[positions - 1], target_dates) # passes
netkit.assert_windows_past_only(changes.index[positions], target_dates)
# AssertionError: 1980 of 1980 windows end on or after the date they are scored on
```

**make\_windows: all 1,980 windows end the trading day before their target date.**

# Every sequence model so far

| Model               | Ch. | MAE, bp | Below |
|---------------------|-----|---------|-------|
| zero_change         | 1   | 3.21    | 0     |
| training_mean       | 1   | 3.22    | 0     |
| dense_16_changes    | 1   | 3.35    | 0     |
| simple_rnn_16       | 1   | 3.31    | 0     |
| lstm_16             | 1   | 3.23    | 0     |
| gru_16              | 1   | 3.22    | 1     |
| conv1d_16           | 1   | 3.24    | 0     |
| dense_16_11_tenors  | 11  | 3.87    | 0     |
| lstm_16_11_tenors   | 11  | 3.23    | 1     |
| conv1d_16_11_tenors | 11  | 3.26    | 0     |



Zero change is the lowest line; no network beat it on average.

# The choice for day 14

**3.22 bp**

gru\_16, 10 Yr alone: lowest mean MAE

**3.23 bp**

lstm\_16\_11\_tenors: lowest on 11 tenors

**3.21 bp**

zero change, beside them on day 14

## The rule, set before the runs

On each kind of input, the lowest mean walk-forward MAE over the five development folds. Chosen on development dates only; no held-out date fitted or scored.

**Chosen is not good: neither beats zero change, and the seed spread is day 14's.**

# Excel side by side

## Excel side by side (tested in Excel)

`=SUMPRODUCT(B2:B6,$K$2:$K$6)+$K$7`

`=SUMPRODUCT(B2:B6,$M$2:$M$6) = =AVERAGE(B2:B6)`

`=MAX(0,C2)`

`=AVERAGE(D2:D17)` and `=MAX(D2:D17)`

`=SUMPRODUCT(B2:L6,$N$2:$X$6)+$N$7`

C2 to C17: filter 1, first 5.6165

five weights of 0.2: a moving average

ReLU, filled down

pooling: 2.8086 and 9.5347

11 tenors: a 5 by 11 rectangle

**Equal to netkit within 1e-12 and to Keras within 1e-5. Excel cannot fit the weights, and has no walk-forward routine.**

# Save the day: the sequence log

## 1. Morning

Files as day 12 left them

## 2. Code

```
conv1d_by_hand,  
assert_windows_past_only
```

## 3. Log, rebuilt

Days 11 and 12, no levels row

## 4. Today's rows

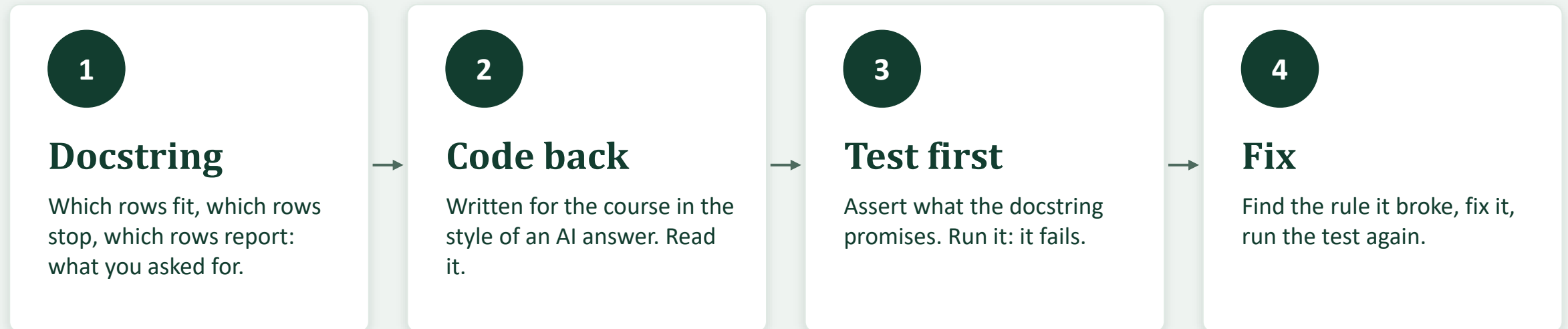
Conv1D, and 11-tenor networks

```
!git -C "{work}" log --oneline --decorate  
# (HEAD -> branch) marks the newest commit; day 14 adds a tag
```

**Columns: model, channels, window, rows, folds, mae\_bp, zero\_change\_mae\_bp, folds\_below\_zero\_change. 56 tests pass.**

# The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: 11-tenor windows end on the target date: fold 1 2.09 bp against zero change's 2.72 bp.

*Code that runs is not the same as code that is right.*

# What you can do now

- Compute a convolution by hand; check it against Keras
- Write a filter in Excel as SUMPRODUCT filled down
- Pool a filter's outputs: average and max
- Score a Conv1D walk-forward against zero change
- Keep same-day changes out of every window

**Week 3, Day 4** Case study: the held-out years, opened once